



Q1 Sum the given number upto single digit

5 marks

Write a Python code to find the sum of the given number upto single digit

input: 89789

output : 5

Your Code

```
1 n = int(input("Enter a number: "))
2
3 while n >= 10:
4     s = 0
5     while n > 0:
6         s = s + (n % 10)
7         n = n // 10
8     n = s
9
10 print("Output:", n)
```

Input

89789

Output

Enter a number: Output: 5



Run



Save



Submit



Q3 Swap the first and last digits

5 marks

WAP to swap the first and last digits of the given number.

Input: 23546

Output: 63542

Your Code

```
1 n = input("Enter a number: ")
2
3 first = n[0]
4 last = n[-1]
5 middle = n[1:-1]
6
7 result = last + middle + first
8
9 print("Output:", result)
```

Input

23546

Output

Enter a number: Output: 63542



Run



Save



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WAP to check the given number is step number or not

A step number is a number in which the difference between every pair of consecutive digits is 1.

For example:

54567 → Yes

$|5-4| = 1$

$|4-5| = 1$

$|5-6| = 1$

$|6-7| = 1$

1235 → No

$|3-5| = 2$

Your Code

```
1 n = input("Enter a number: ")
2
3 flag = 1
4
5 for i in range(len(n) - 1):
6     if abs(int(n[i]) - int(n[i + 1])) != 1:
7         flag = 0
8         break
9
10 if flag == 1:
11     print("Yes")
12 else:
13     print("No")
```

Input

54567
1235

Output

Enter a number: Yes

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Q4 Odd Sequence first and Even Sequence next

5 marks

WAP to odd Sequence first and Even Sequence next

Input: 658134

Output: 513684

Your Code

```
1 n = input("Enter a number: ")
2
3 odd = ""
4 even = ""
5
6 for i in range(len(n)):
7     if i % 2 == 0:
8         odd = odd + n[i]
9     else:
10        even = even + n[i]
11
12 print("Output:", odd + even)
```

Input

658134

Output

Enter a number: Output: 683514



Run



Save



Submit



Q6 Reverse only the even numbers

5 marks

WAP to reverse only the even numbers

Input: 325649

Output: 345629

Your Code

```
1 n = input("Enter a number: ")
2
3 even = ""
4
5 for i in n:
6     if int(i) % 2 == 0:
7         even = i + even
8
9 result = ""
10
11 for i in n:
12     if int(i) % 2 == 0:
13         result = result + even[0]
14         even = even[1:]
15     else:
16         result = result + i
17
18 print("Output:", result)
```

Input

325649

Output

Enter a number: Output: 345629



Q2 Prime Number

5 marks

Find the next prime number

input: 13

output: 17

Your Code

```
1 n = int(input("Enter a number: "))
2
3 n = n + 1
4
5 while True:
6     prime = True
7
8     for i in range(2, n):
9         if n % i == 0:
10             prime = False
11             break
12
13     if prime:
14         print("Output:", n)
15         break
16
17     n = n + 1
```

Input

13

Output

Enter a number: Output: 17



Run



Save



Submit



Q7 Encoding

5 marks

WAP to encode the given number

Input: 55553322255

Output: 54322352

Your Code

```
1 n = input("Enter a number: ")
2
3 i = 0
4
5 while i < len(n):
6     count = 1
7
8     while i < len(n) - 1 and n[i] == n[i + 1]:
9         count += 1
10        i += 1
11
12    print(n[i], count, sep="", end="")
13    i += 1
```

Input

55553322255

Output

Enter a number: 54322352



Q8 Reverse Odd Digit

5 marks

WAP to Reverse Odd Digits Only

Input: 816253

Output: 351

Your Code

```
1 n = input("Enter a number: ")
2
3 odd = ""
4
5 for i in n:
6     if int(i) % 2 != 0:
7         odd = i + odd
8
9 print("Output:", odd)
```

Input

816253

Output

Enter a number: Output: 351